01 PROTOCOL

PRIVATE RECURRING CRYPTO SUBSCRIPTIONS FOR SOLANA

v0.9.9 · FROST RELEASE

Volta Team · Slashy Fx · Solo, full-time

Dogpatch · Superteam · April 2026

Crypto rails leak. Card rails take 12%.

A merchant accepting subscriptions today picks one of two losing trades:

Public chains

Solana Pay, Coinbase Commerce, every standard L1 rail.

Every customer's wallet history is on display.
The merchant sees it.
The chain sees it.
Anyone querying the chain sees it.

Card processors

Stripe, PayPal, Patreon, Apple / Google in-app.

2.9% to 12% rake.
The processor surveils every transaction.
Chargebacks are a loaded gun the buyer holds.

No recurring privacy rail

Mixers (Tornado), shielded pools (Aztec, Zcash).

Privacy primitives exist but no merchant SDK, no recurring billing, no Solana speed.
Built for transfers, not subscriptions.

Three lines crossed in 18 months.

2024 Q3

Compute budget bump

Solana raised the per-tx CU ceiling.
On-chain STARK verification became economical (~900K CU, <0.001 SOL per verify).
Custom FRI verifiers in 792 KB SBF binaries became feasible.

2024 AUG

ML-KEM-768 finalised

NIST published the post-quantum KEM standard.
Hybrid X25519 + ML-KEM-768 stealth addresses are now production-grade.
Privacy that survives a future quantum adversary.

2024-2026

Creator-rail revolt

Patreon's 12% rake + Apple's 30% in-app + Stripe 2.9%.
Creators publicly leaving for crypto rails (Pump.fun, Hivemapper, indie SaaS).
The market is asking for an exit.

The cryptography is ready. The chain is ready. The market is ready.

Private recurring crypto subscriptions.

Drop-in SDK for any online merchant.
Customer subscribes from the Protocol 01 wallet (or via QR scan).
Merchant receives SOL or USDC at their own retailer wallet.
Every link between the two is severed by ZK-STARK proofs and hybrid post-quantum stealth addresses.

0.80%
TOTAL ROUND-TRIP FEE

0%
MERCHANT PLATFORM CUT

~10s
ON-CHAIN SETTLEMENT

PQ
POST-QUANTUM HARDENED

Three integration paths

`P01-JS`

Drop-in pay button + React widgets. 5 lines of code on the merchant's site.

`MERCHANT-SDK`

Server-side: register service, poll payments, watch vaults, issue access tokens.

`AUTH-SDK`

"Login with Protocol 01".
QR + biometric, gate by active subscription.

Functional prototype, on-chain. Not yet adopted.

The rail is real and end-to-end.
No external merchant has integrated yet, and that is the honest stage we are at.
The cryptography and the SDK are ahead of the adoption curve.

What is proven, today

- 14 Anchor programs deployed on devnet · 6 STARK AIRs in a custom 792 KB FRI verifier
- Registry program live: `QaQwpvBi1EQpevNE21D2oNBHFsLtoLwa7aXH26zRhQB`
- 22/22 live-devnet E2E checks: register → fetch → deregister, every field byte-equal, rent refunded
- 2,400+ tests passing across 17 SDKs · 26/26 fresh-project tarball install
- Mobile (Android), Chrome extension, web — all three clients shipped

```
// real on-chain signatures (devnet)
register   3odKFeJXyvpVcy7q6Df5tzqM627DE6gt5...
deregister 4MPXSxxRyocawcGuYg93aWUVKj9WPvyx2...
```

What is NOT proven, yet

- **External merchant adoption.**
  No real online business has integrated Protocol 01 today.
- The five entries seeded on the devnet registry (`netflix-standard`, `disney-plus`, etc.) are **demo placeholders** we created ourselves to validate the registry mechanism.
  Not partnerships.
- That crypto-native recurring privacy is a market need **today**.
  Web3 has not reached the threshold where every merchant must adopt it.

The bet.
Build the rail before the wave.
When privacy + post-quantum + creator-fee fatigue cross, the merchants who need it will find an SDK that has been hardened for 12 months.
Not one being written that day.

Five lines on the merchant's server.

Frontend ships a drop-in widget.
Backend runs a 30-second poll loop.
That's it.

Frontend — React

```
import { P01Provider, SubscriptionWidget }
  from '@protocol-01/p01-js/react';

<P01Provider config={{ merchantId: 'acme' }}>
  <SubscriptionWidget tiers={[
    { id: 'pro', price: 19.99, interval: 'monthly' }
  ]} />
</P01Provider>
```

Backend — Node / Cloudflare

```
import { pollPaymentsForRetailer, issueAccessToken }
  from '@protocol-01/merchant-sdk';

setInterval(async () => {
  const r = await pollPaymentsForRetailer(c, retailer);
  r.forEach(p => grantAccess(p));
}, 30_000);
```

What the merchant never sees

- HIDDEN The customer's wallet identity
- HIDDEN The customer's transaction history
- HIDDEN Any link between two subscribers

14 programs. 1 custom STARK verifier.

Our deployed programs

VERIFIER	p01_stark_verifier — FRI, 6 AIRs
CORE	zk_shielded — pool + sub vault
CORE	specter — stealth + streams
CORE	p01_registry — merchant directory
CORE	p01_zkspl — confidential SPL
PQ	p01_quantum_vault — WOTS+ + hash-lock
PQ	p01_arcium — Cerberus MPC bridge
RAILS	p01_relayer — trustless relay
RAILS	p01_liquidity — instant unshield
RAILS	subscription / stream / fee-splitter
RAILS	p01_mugen — P2P fiat escrow

Solana primitives leveraged

Anchor 0.32	DSL/framework, all 14 programs
SPL Token	USDC + native SOL handling
PDAs	deterministic registry / vault keys
Compute Budget	1.4M CU for STARK verify
SystemProgram	native SOL + account allocation
Memo Program v2	invoice memos for one-shots
Helius RPC	getProgramAccounts pagination
Privy	embedded wallet onboarding
Arcium Cerberus	9 MPC circuits, threshold compute

The custom FRI/STARK verifier fits in a **792 KB** SBF binary. ~900K CU per verification. No other L1 hosts a verifier of this scope at this cost.

Built today, in 90 days, alone.

14	17	6	2,400+
ANCHOR PROGRAMS	SDKS PUBLISHED	STARK AIRS	TESTS PASSING

Shipped

- 14 Anchor programs deployed on devnet, ~340 program tests
- Custom FRI/STARK verifier (792 KB SBF, 6 circuits)
- Mobile app v0.9.9 Frost (Android APK 119 MB)
- Chrome extension MV3 with on-device WASM prover
- Web marketing site · design doc · economic charter
- Mugen reference fiat <> crypto P2P (FROST escrow)
- 5 verified merchants on devnet registry

Left before May 11

- External audit cycle (OtterSec / Neodyme — quote in flight)
- Mainnet deployment + first 5 paying merchants
- iOS TestFlight beta
- Atomic dust-to-stealth in `cancel_private_stark`

Biggest risk

Audit timing.
Privacy primitives without an audit are theatre.
We do not deploy to mainnet without one full external cycle complete.

0.80% round-trip. Zero platform cut.

0.30%	0.50%	0.80%
SHIELD FEE	UNSHIELD FEE	ROUND-TRIP TOTAL

Versus the rails merchants use today

PROVIDER	TAKE RATE	PRIVACY	SETTLEMENT
PROTOCOL 01	0.80%	NATIVE ZK	~10s on-chain
Stripe (cards)	2.90%	NONE	2-7 days
Patreon	8-12%	NONE	Monthly
Coinbase Commerce	1.00%	NONE	~30 min finality
BTCPay self-host	0.00%	PSEUDONYMOUS	10-60 min
Solana Pay (raw)	0.00%	NONE	~1s

Worked example: SaaS at €19.99/mo · 1,000 subs · 12 months · 10% churn · gross €143,449 → net to merchant **€142,668**. Same revenue on Stripe: ~€5,460 fee · on Patreon: €14,345 to €17,214 fee.

Built by hand.

Cryptography I implemented from first principles.
Not glued together from libraries:

- Custom on-chain **FRI/STARK verifier** (Goldilocks field, Poseidon AIR, 6 circuits, 792 KB SBF binary)
- Hybrid **X25519 + ML-KEM-768** stealth address construction (own design, not a reference port)
- **WOTS+** post-quantum signature scheme + SHA-256 hash-timelock vault
- Deterministic **HKDF note-secret derivation** — one seed, every shielded note recoverable cross-device
- Custom **MPC bridge to Arcium Cerberus** — 9 Arcis circuits, threshold compute
- **14 Anchor programs**, 17 SDKs, 2,400+ tests — all written by the same person

> ***"Financial privacy is a right, not a feature. No team, no funding, just execution."***
>
> — Slashy, founder

The CV is on the form.
The code is the credentials.

What we need to ship.

The core is built.
The cryptography is proven.
The chain is ready.
What stands between Protocol 01 and 100 paying merchants on mainnet:

Audit funding

One full external cycle from OtterSec, Neodyme or Trail of Bits. Quote pending. ~$80K-150K.

First merchant cohort

5 to 10 Solana-native creators or SaaS to seed the verified registry on mainnet. Friend-of-Solana intros highest-leverage.

Ecosystem partnerships

Helius (RPC), Privy (wallet onboarding), Arcium (MPC), Backpack (distribution). Conversations in flight.

Privacy is a right.
Recurring is the rail.
Solana is the chain.

PROTOCOL-01.DEV · @PROTOCOL01_

VOLTA TEAM · SLASHY FX · SOLO, FULL-TIME

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